

Experiment No. 26

Title

Prolog Program for Fruit and Its Color using Backtracking

Aim

To write a Prolog program to find the **color of a fruit using backtracking**.

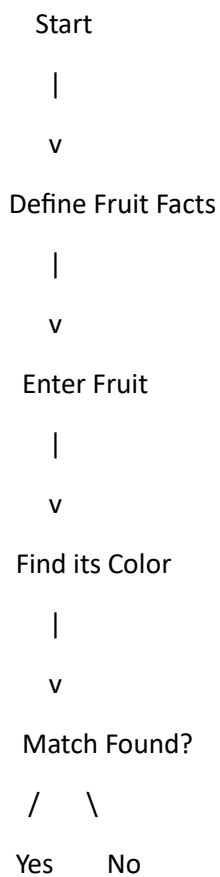
Objective

To understand how Prolog uses facts and backtracking to find matching solutions.

Algorithm

1. Start.
2. Define facts for fruits and their colors.
3. Enter a fruit as a query.
4. Search for its matching color.
5. Use backtracking to find other matches if available.
6. Display the result.
7. Stop.

Flowchart



| |
v v

Display Color Not Found

|
v

More Matches?

/ \

Yes No

| |

Backtrack |

|_____|
|
v

Stop

Prolog Program

color(apple, red).

color(apple, green).

color(banana, yellow).

color(orange, orange).

color(grapes, green).

color(mango, yellow).

find_color(Fruit, Color) :-

color(Fruit, Color).

Queries

?- find_color(apple, Color).

Sample Output

Color = red ;

Color = green.

Another query:

?- find_color(banana, Color).

Output:

Color = yellow.

Result

The Prolog program successfully found the color of a given fruit using **backtracking**.

Conclusion

Thus, fruit and color information was successfully represented and retrieved using Prolog backtracking.